Electrical Load Charts – General Data

1. General

The charts given in this section relate to loading of a typical aircraft electrical system.

Table 1 28 Volt DC Load Chart – Left Bus

CB #	CB Grid	Load Equipment Left Bus	CB Rating (Amps)	Rated Current (Amps)
105	**A-03**	GEN FAIL R	5.00	0.20
107	**A-04**	CABIN HEAT	5.00	0.45
109	**A-05**	WINDSHIELD WIPERS	10.00	4.80
111	**A-06**	PITOT HT LEFT AND STALL SENSOR	7.50	7.30
113	**A-07**	PY (FUEL CONT SENSOR) HEATER LEFT	5.00	3.90
115	**A-08**	FUEL PUMP FWD #1	15.00	6.50
117	**A-09**	FUEL PUMP AFT #1	15.00	6.50
119	**A-10**	BOOST PUMP PRESSURE 1	5.00	0.04
121	**A-11**	TORQUE PRESS LEFT	5.00	0.01
123	**A-12**	OIL PRESS IND LEFT	5.00	0.01
125	**A-13**	WING FUEL CONTROLS L	7.50	5.80
127	**A-14**	FLIGHT CONTROLS INDICATION	5.00	0.05
	A-15	RESERVED		
131	**A-16**	MAP LIGHTS & PASS WARN SIGN LTS	5.00	0.22
133	**A-17**	CABIN LTS BRIGHT	5.00	0.93
	A-18	RESERVED		
137	**C-01**	WINDSHIELD HEAT L	5.00	0.36
1	**NA**	WINDSHIELD HEAT L	30.00	26.00
141	**C-03**	ENGINE IGNITORS LEFT	5.00	1.20
143	**C-04**	VALVE BLEED AIR LEFT	5.00	0.76
145	**C-05**	PROPELLER DEICE	20.00	18.00
147	**C-06**	HYD OIL PUMP (INCLUDES K11 RELAY)	35.00	30.00
149	**C-07**	PROPELLER AUTOFEATHER	5.00	0.10
151	**C-08**	FUEL SOV LEFT	5.00	
153	**C-09**	FUEL QTY IND AFT/LEFT	5.00	0.03
155	**C-10**	FIRE EXT BOTTLE	5.00	
157	**C-11**	FIRE DETECTION LEFT	5.00	0.01
	C-12	RESERVED		
161	**C-13**	FIRE DET FAULT IND L	5.00	0.01

Table 1 28 Volt DC Load Chart – Left Bus (continued)

CB #	CB Grid	Load Equipment Left Bus	CB Rating (Amps)	Rated Current (Amps)
	C-14	RESERVED		
165	**C-15**	BEACON LT (PRE MOD 6/2240)	5.00	0.19
165	**C-15**	BEACON LT (POST MOD 6/2240)	5.00	0.67
167	**C-16**	LANDING LT LEFT (PRE MOD 6/2285)	7.50	2.20
167	**C-16**	LANDING LT LEFT (POST MOD 6/2285)	5.00	1.50
169	**C-17**	WING POS LTS (INCL TAIL POS LT)	5.00	0.52
171	**C-18**	AIRFRAME DEICE AUTO	5.00	3.20
201	**E-01**	MAU POWER SUPPLY MODULE 1A	20.00	1.30
221	**E-03**	AUDIO/MARKER PANEL PILOT	5.00	1.53
241	**E-05**	DATA ACQUISITION UNIT 1A	5.00	0.30
261	**E-07**	AVIONICS FAN	1.00	0.22
203	**F-01**	MAU POWER SUPPLY MODULE 2A	20.00	1.30
223	**F-03**	MULTI-MODE DIGITAL RADIO 1	7.50	4.40
243	**F-05**	DATA ACQUISITION UNIT 2A	5.00	0.40
	F-07	RESERVED (VIDEO)		
205	**G-01**	AURAL ALERT 1	1.00	0.01
225	**G-03**	MODE S APEX TRANSPONDER 1	5.00	1.64
245	**G-05**	ENHANCED GROUND PROXIMITY COMPUTER	3.00	0.28
	G-07	RESERVED (SATELLITE TRACKING)		
207	**H-01**	PILOT PFD, FLAT PANEL DISPLAY, 10.4” DIA	5.00	0.50
227	**H-03**	DME 1	3.00	0.55
247	**H-05**	TCAS I	5.00	1.65
247	**H-05**	TCAS II	5.00	1.02
	H-06	RESERVED		
267	**H-07**	GEAR INDICATION	5.00	0.04
209	**J-01**	UPPER MFD, FLAT PANEL DISPLAY, 10.4” DIA	5.00	0.50
229	**J-03**	GPS SENSOR UNIT 1	3.00	0.25
249	**J-05**	RADAR ALTIMETER 1	3.00	0.50
	J-07	RESERVED (GEAR ADVISORY)		
211	**K-01**	ADAHRS 1A	3.00	0.36
231	**K-03**	AR-120 CVR SYSTEM	5.00	0.28
251	**K-05**	C406–N ELT NAV INTERFACE	1.00	0.06
	K-07	RESERVED		

Table 1 28 Volt DC Load Chart – Left Bus (continued)

CB #	CB Grid	Load Equipment Left Bus	CB Rating (Amps)	Rated Current (Amps)
213	**M-01**	FLIGHT CONTROLLER 1A	3.00	0.09
233	**M-03**	AR-256 FDR SYSTEM (PRE MOD 6/2332)	5.00	0.11
233	**M-03**	FA2100 FDR SYSTEM (512) (POST MOD 6/2332)	5.00	0.34
253	**M-05**	NAT 245 CABIN PA AMPLIFIER SYSTEM	2.50	0.60
273	**M-07**	AVIONICS STRT BATT – CHARGE (POST MOD 6/2315)	30.00	2.00
215	**N-01**	PITCH TRIM (S.O.O. 6221)	5.00	1.10
235	**N-03**	CAUTION LIGHTS NORMAL	2.00	0.26
255	**N-05**	ESIS	15.00	5.47
275	**N-07**	AVIONICS STRT BATT – CTRL (POST MOD 6/2315)	2.00	0.35
NA	**NA**	GENERATOR FIELD RELAY		0.35
NA	**NA**	GENERATOR RELAY		0.40
NA	**NA**	BATTERY RELAY		0.40
NA	**NA**	GENERATOR OVERVOLT RELAY		0.16
NA	**NA**	GENERATOR REVERSE CURRENT RELAY		0.37
NA	**NA**	MAIN BATT CHARGE		48.00
217	**P-01**	AP SERVOS (S.O.O. 6221)	7.50	2.00
216	**P-02**	RESERVED (CCD CONTROL)		
237	**P-03**	WX RADAR	5.00	2.04
257	**P-05**	ESIS INSTRUMENT	3.00	0.00
277	**P-07**	PANEL LIGHTS L	5.00	0.69
219	**Q-01**	DISPLAY CONTROLLER PILOT	3.00	0.13
239	**Q-03**	MFD CONTROLLER	3.00	0.20
259	**Q-05**	ESIS MAGNETOMETER	1.00	0.00
279	**Q-07**	FLOOD LIGHTS	5.00	0.50
21	**R-01**	BAGGAGE COMP LT FWD	5.00	0.11
21	**R-1**	BAGGAGE COMP LT REAR		0.22
21	**R-01**	ENTRANCE LT		0.11
26	**R-02**	POWER AMP (HF COMMUNICATION SYSTEM)	30.00	13.00
	R-03	RESERVED		
22	**S-01**	DOME LIGHT AND CLOCK	5.00	0.24
25	**S-02**	RX/EX ANT (HF COMMUNICATION SYSTEM)	5.00	4.00
	S-03	RESERVED		
23	**T-01**	BATTERY VOLTIMETER	1.00	0.01

Table 1 28 Volt DC Load Chart – Left Bus (continued)

CB #	CB Grid	Load Equipment Left Bus	CB Rating (Amps)	Rated Current (Amps)
27	**T-02**	EMERGENCY LIGHTS BATTERY	10.00	2.00
	T-03	RESERVED		
24	**U-01**	ELT SWITCH	1.00	0.00
28	**U-02**	DAU COOLING FAN	2.00	0.40
	U-03	RESERVED		

Table 2 28 Volt DC Load Chart – Right Bus

CB #	CB Grid	Load Equipment Right Bus	CB Rating (Amps)	Rated Current (Amps)
	B-01	RESERVED		
104	**B-03**	GEN FAIL L	5.00	0.20
106	**B-04**	VENT FAN	20.00	18.00
108	**B-05**	INTAKE DEFLECTORS	10.00	5.30
110	**B-06**	PITOT HT RIGHT	7.50	5.30
112	**B-07**	PY (FUEL CONT SENSOR) HEATER RIGHT	5.00	3.92
114	**B-08**	FUEL PUMP FWD #2	15.00	6.05
116	**B-09**	FUEL PUMP AFT #2	15.00	6.24
118	**B-10**	BOOST PUMP PRESSURE 2	5.00	0.04
120	**B-11**	TORQUE PRESS IND RIGHT	5.00	0.01
122	**B-12**	OIL PRESS IND RIGHT	5.00	0.01
124	**B-13**	WING FUEL CONTROLS R	7.50	5.80
126	**B-14**	TAXI LIGHT R (S.O.O. 6441)	5.00	3.60
126	**B-14**	TAXI LIGHT R (S.O.O. 6442)	5.00	1.50
128	**B-15**	READING LIGHTS	5.00	1.10
130	**B-16**	TAXI LIGHT	5.00	1.50
130	**B-16**	TAXI LIGHT (S.O.O. 6440)	5.00	3.00
130	**B-16**	TAXI LIGHT L (S.O.O. 6441)	5.00	3.60
130	**B-16**	TAXI LIGHT L (S.O.O. 6442)	5.00	1.50
132	**B-17**	DOORS UNLOCKED	5.00	0.16
134	**B-18**	ACTUATOR AILERON TRIM	5.00	0.10
136	**D-01**	WINDSHIELD HEAT R	5.00	0.36
2	**NA**	WINDSHIELD HEAT R	30.00	26.00
18	**NA**	STN 332 CABIN 28V DC PLUG (S.O.O. 6283)	5.00	5.00

Table 2 28 Volt DC Load Chart – Right Bus (continued)

CB #	CB Grid	Load Equipment Right Bus	CB Rating (Amps)	Rated Current (Amps)
18	**NA**	STN 332 CABIN 14V DC PLUG (S.O.O. 6272)	5.00	5.00
140	**D-03**	ENGINE IGNITOR RIGHT	5.00	1.26
142	**D-04**	VALVE BLEED AIR RIGHT	5.00	1.00
144	**D-05**	PROPELLER OVERSPEED GOVERNOR	5.00	0.76
146	**D-06**	HYDRAULIC PRESSURE INDICATOR	5.00	0.02
148	**D-07**	FUEL CROSS-FEED SOV	5.00	2.01
150	**D-08**	FUEL SOV RIGHT	5.00	0.00
152	**D-09**	FUEL QTY IND FWD/RIGHT	5.00	0.03
154	**D-10**	FIRE EXT BOTTLE	5.00	0.00
156	**D-11**	FIRE DETECTION RIGHT	5.00	0.01
	D-12	RESERVED		
158	**D-13**	FIRE DET FAULT IND R	5.00	0.00
	D-14	RESERVED		
164	**D-15**	STROBE LIGHTS	5.00	2.30
166	**D-16**	LANDING LIGHT RIGHT (PRE MOD 6/2285)	7.50	2.20
166	**D-16**	LANDING LIGHT RIGHT (POST MOD 6/2285)	5.00	1.25
168	**D-17**	WING INSPECTION LIGHT	5.00	0.44
170	**D-18**	AIRFRAME DEICE MANUAL	5.00	0.00
200	**E-02**	MAU POWER SUPPLY MODULE 1B	20.00	1.60
220	**E-04**	AUDIO/MARKER PANEL COPILOT	5.00	0.33
240	**E-06**	DATA ACQUISITION UNIT 1B	5.00	0.29
202	**F-02**	MAU POWER SUPPLY 2B	20.00	1.60
222	**F-04**	MULTI-MODE DIGITAL RADIO 2	7.50	4.4
242	**F-06**	DATA ACQUISITION UNIT 2B	5.00	0.4
204	**G-02**	AURAL ALERT 2	1.00	0.01
224	**G-04**	MODE S APEX TRANSPONDER 2	5.00	1.64
244	**G-06**	HEADSETS	2.00	0.05
	H-06	RESERVED		
206	**H-02**	COPILOT PFD, FLAT PANEL DISPLAY, 10.4” DIA	5.00	0.50
226	**H-04**	DME 2	3.00	0.55
208	**J-02**	LOWER MFD, FLAT PANEL DISPLAY, 10.4” DIA	5.00	0.50
228	**J-04**	GPS SENSOR UNIT 2	3.00	0.25
248	**J-06**	RADAR ALTIMETER 2	3.00	2.0

Table 2 28 Volt DC Load Chart – Right Bus (continued)

CB #	CB Grid	Load Equipment Right Bus	CB Rating (Amps)	Rated Current (Amps)
210	**K-02**	ADAHRS 1B	3.00	0.36
230	**K-04**	SATELLITE WEATHER	5.00	0.22
250	**K-06**	RESERVED (AUTO BRIEF)		
212	**M-02**	FLIGHT CONTROLLER 1B	3.00	0.1
232	**M-04**	RESERVED (STORMSCOPE OPTION)	3.00	0.22
252	**M-06**	PWR CONV 28V INPUT	5.00	0.04
214	**N-02**	CAUTION LTS BYPASS	2.00	
234	**N-04**	SAT COM (S.O.O. 6298)	3.00	0.50 CONT. 0.80 MAX
236	**P-04**	HF COMMUNICATION SYSTEM CTRL	2.00	0.12
	P-06	RESERVED		
218	**Q-02**	DISPLAY CONTROLLER COPILOT	3.00	0.13
238	**Q-04**	ECTM/FOQA (S.O.O. 6288)	3.00	0.15
258	**Q-06**	VHF-FM (S.O.O. 6296)	3.00	0.70